

1. General Description

The EMS4321 is a high performance, Double-Pole, Double Throw (DPDT) multiplexer that combines a low-distortion audio and a USB2.0 High-Speed (HS) switch path. This configuration enables audio and USB data to share a common connector port. The architecture is designed to allow audio signals to swing below ground. This means analog audio over USB-C connector can be used for personal media players and portable peripheral devices. With Max. 12V high voltage range of VBUS pin, EMS4321 can be configured for automatic USB communication with VBUS tie to VBUS directly. The EMS4321 includes an over voltage protection of D+/R & D-/L pin to protect internal port. Typical applications involve switching in portables and consumer applications such as mobile phone, tablet, laptop, portable media players. The EMS4321 is available in QFN1.4×1.8 package. Standard products are Pb-Free and halogen-Free.

2. Features and Benefits

- Single Supply Range: 2.5V to 5.5V
- VBUS Pin Voltage: 13V
- D+/R & D-/L Pin OVP Threshold: 4.8V
- Audio Path On-Resistance: 0.9Ω
- USB Path On-Resistance: 3.4Ω
- USB Path -3dB Bandwidth: 1000MHz
- THD+N@ 32Ω load & 0.7V_{rms}: -105dB
- Signal-to-Noise (SNR) Ratio: -115dB
- Audio Swing Range (R_L=1KΩ): -2.5V to +2.5V

EMS4321**USB2.0 Hi-Speed and HiFi Audio/UART Switches with Negative Swing Capability**

3.Applications

- USB Type-C Audio

4.Ordering Information

Table 1. Ordering information

Type number	Package		
	Name	Description	Quantity
EMS4321RSW	QFN1.4×1.8-10L	Quad Flat No-leads Package, 10 pins; body width 1.4 ×1.8mm; 0.4 mm pitch, 0.6 mm(max) height	3000

5. Pinning Information

5.1. Pinning

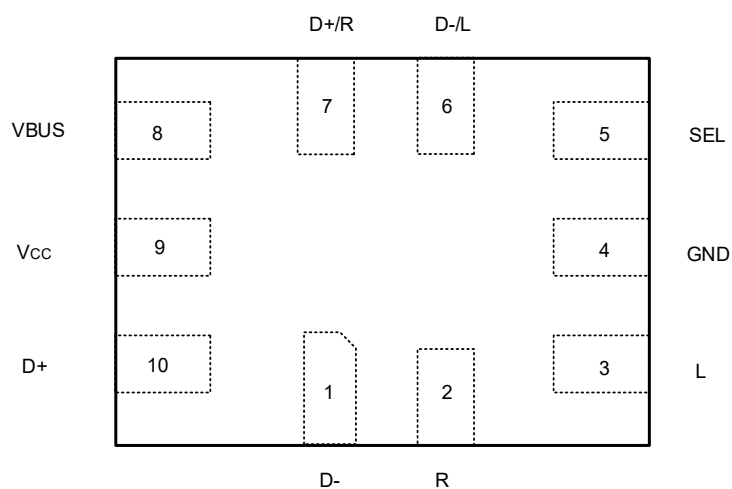


Fig. 1. Top view pin configuration QFN 1.4x1.8-10L

5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
D-	1	USB data (Negative) input sources
R	2	Audio Right input sources
L	3	Audio Left input sources
GND	4	Ground
SEL	5	Audio Selection and Low-Power Mode Control
D-/L	6	USB data (Negative) and audio Left common connector ports
D+/R	7	USB data (Positive) and audio Right common connector ports
VBUS	8	USB Switch Selection
V _{CC}	9	Power supply
D+	10	USB data (Positive) input sources

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

VBUS	SEL	USB Mode	Audio Mode
H	L	ON	OFF [1]
L	L	OFF	ON
H	H	OFF	ON
L	H	OFF	OFF [1]

[1] 70Ω R_T resistors are enabled in this state.

7. Typical Application

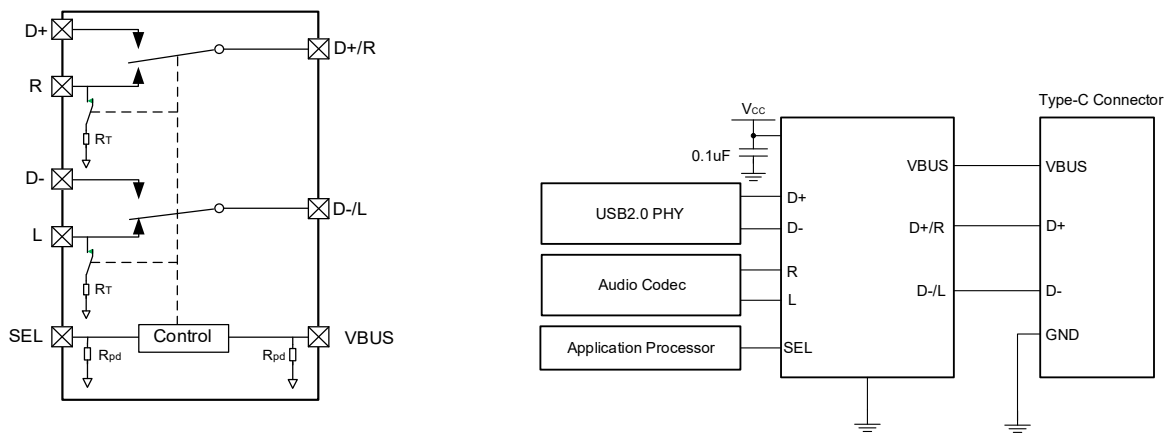


Fig. 2 USB2.0 Hi-Speed and HiFi Audio/UART Switches

8. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Min	Max	Unit
V_{CC}	Supply voltage range	-0.3	6.0	V
V_{D+}/V_{D-}	USB signal voltage range	-0.3	5.5	V
V_R/V_L	Audio signal voltage range	-3.0	5.5	V
V_{VBUS}	VBUS pin voltage range	-0.3	12	V
V_{SEL}	SEL pin voltage range	-0.3	6.0	V
I_{OUT}	Continues output current (USB)	-100	100	mA
I_{OUT}	Continues output current (Audio)	-200	200	mA
T_J	Junction temperature range		150	°C
T_L	Lead temperature range		260	°C
T_{STG}	Storage temperature range	-65	150	°C
$R_{\theta JA}$	Thermal resistance		250	°C/W
ESD	Human Body Model (HBM)	-3000	3000	V

9. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Min	Max	Unit
V _{CC}	Supply voltage range	2.5	5.5	V
V _{USB}	USB signal voltage range	0	4.0	V
V _{AUDIO}	Audio signal voltage range	-2.5	2.5	V
V _{VBUS}	VBUS pin voltage range	0	12	V
V _{SEL}	SEL pin voltage range	0	V _{CC}	V
T _{OPR}	Operating temperature range	-40	85	°C

10. Electrical Characteristics

Table 6. Electrical characteristics (Ta = 25 °C, V_{CC} = 3.3 V, unless otherwise specified.)

Symbol	Parameter	Conditions	Min.	Typ.	Max.	UNIT
V _{IH}	control input logic high	V _{CC} = 2.5 to 3 V	1.3			V
		V _{CC} = 3 to 3.5 V	1.5			
V _{IL}	control input logic low	V _{CC} = 2.5 to 3 V			0.3	V
		V _{CC} = 3 to 3.5 V			0.5	V
I _{CC-A}	Quiescent Supply Current	Audio Mode ON		66		uA
I _{CC-U}		USB Mode ON		35		uA
I _{CC_OFF}		Audio Mode OFF and USB Mode OFF		35		uA
I _{IN}	SEL Input Leakage Current	0 < V _{IS} < V _{CC}			2	uA
I _{IN}	VBUS Input Leakage Current	0 < V _{IS} < 20 V			2	uA
I _{D+(OFF)} I _{D-(OFF)}	D+, D- Off State Leakage Current	VBUS = 0 V, SEL = 3.3 V; V _{D+} , V _{D-} = 0.3 V; V _{D+/R} , V _{D-/L} = 0.3 V;			2	uA
I _{R(OFF)} I _{L(OFF)}	R, L Off State Leakage Current	VBUS = 3.3 V, SEL = 0 V; V _{D+/R} , V _{D-/L} = 3.3 V; V _L , V _R = float;			2	uA
R _{PD}	SEL and VBUS Internal Pull-Down Resistors			4		MΩ
R _T	Audio Path Termination Resistors			70		Ω
R _{ONUSB}	USB Switch On Resistance	V _{D+/D-} = 0 to 0.4 V, I _{ON} = 10 mA		3.4	4.5	Ω
ΔR _{ONUSB}	USB Switch Delta R _{ON} [1]	V _{D+/D-} = 0 to 0.4 V, I _{ON} = 10 mA		0.3		Ω
R _{ONAudio}	Audio Switch On Resistance	V _{L/R} = -2 to 2 V, I _{ON} = 60 mA		0.9	1.2	Ω
ΔR _{ONAudio}	Audio Switch Delta R _{ON} [1]	V _{L/R} = -2 to 2 V, I _{ON} = 60 mA		0.02		Ω
R _{FLATAudio}	Audio Switch R _{ON} Flatness [2]	V _{L/R} = -2 to 2 V, I _{ON} = 60 mA		0.02		Ω
X _{TALK}	Non-Adjacent Channel Crosstalk	Audio Mode, f = 1 KHz, R _T = 32 Ω		-80		dB
		USB Mode, f = 240 MHz, R _T = 50 Ω		-35		dB
O _{IRR}	Off Isolation	Audio Mode, f = 1 KHz, R _T = 32 Ω		-100		dB
		USB Mode, f = 240 MHz, R _T = 50 Ω		-40		dB
BW	-3dB Bandwidth (USB Mode)	R _T = 50 Ω, C _L = 0 pF, Signal 0dBm		1000		MHz
THD	Total Harmonic Distortion (Audio Mode)	f = 20 Hz to 20 KHz, R _L = 32 Ω, V _{IN} = 0.7 V _{rms}		-105		dB
		f = 20 Hz to 20 KHz, R _L = 1 KΩ, V _{IN} = 1.5 V _{rms}		-110		dB
SNR	Signal-to-Noise Ratio (Audio Mode)	f = 20 Hz to 20 kHz, R _L = 32 Ω, V _{IN} = 2V _{rms}		-115		dB

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C _{ON}	D+/R, D-/L On Capacitance	V _{Bias} = 0.2 V, f = 1 MHz		5		pF
C _{OFF1}	USB Path Off Capacitance	V _{Bias} = 0.2 V, f = 1 MHz		3		pF
C _{OFF2}	Audio Path Off Capacitance	V _{Bias} = 0.2 V, f = 1 MHz		4		pF
V _{OVP}	D+/R & D-/L OVP Threshold			4.8		V

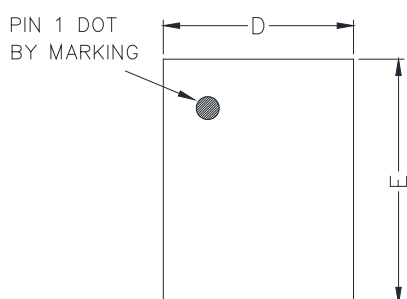
Note:

[1] R_{ON} matching between channels is calculated by subtracting the channel with the lowest max Ron value from the channel with the highest max Ron value.

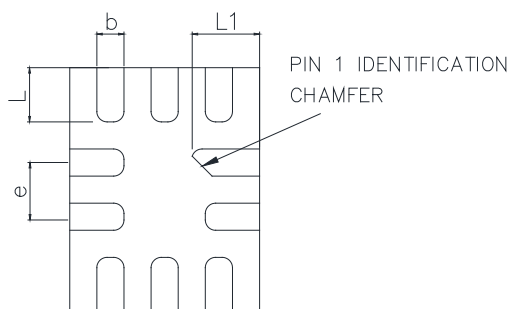
[2] Flatness is defined as the difference between maximum and minimum value of ON-resistance at the specified analog signal voltage points.

11. Package Outline

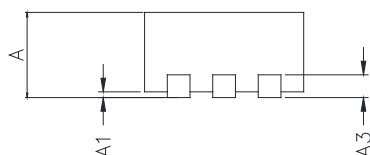
QFN 1.4×1.8-10L



TOP VIEW



BOTTOM VIEW



SIDE VIEW

Symbol	Dimensions in Millimeters		
	Min.	Typ.	Max.
A	0.50	0.55	0.60
A1	0.00	-	0.05
A3	0.15 Ref.		
D	1.35	1.40	1.45
E	1.75	1.80	1.85
b	0.15	0.20	0.25
L	0.30	0.40	0.50
L1	0.40	0.50	0.60
e	0.40 BSC		

12. Tape and Reel Information

12.1. Carrier tape dimensions

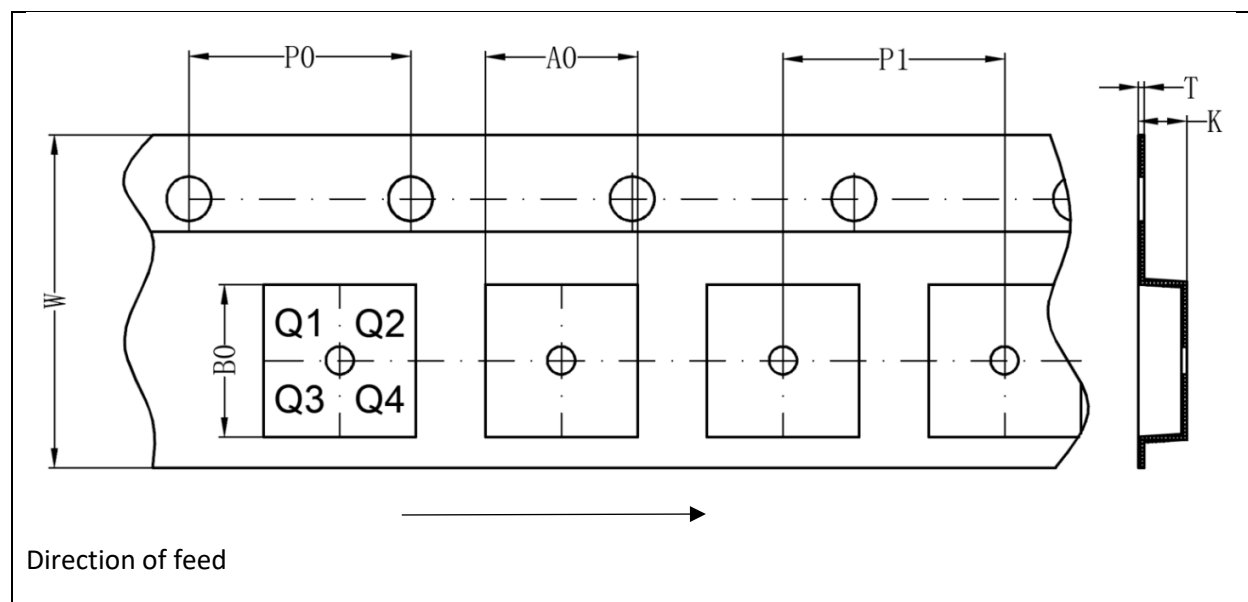


Table 7. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
QFN1.4×1.8-10L	1.6	2.0	0.85	0.2	4	8	4	Q1

12.2. Reel and box dimensions

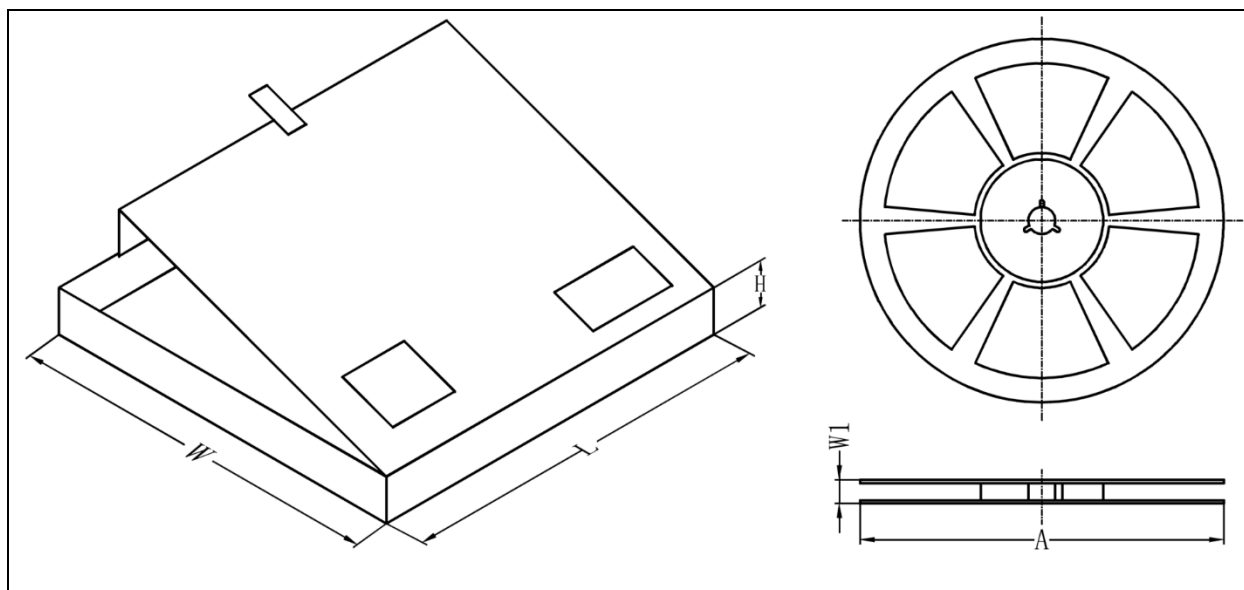


Table 8. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Max Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm)[2]
QFN1.4×1.8-10L	RSW	180	13.2	3000	1	210×200×40

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

13. Abbreviations

Table 9. Abbreviations

Acronym	Description
ESD	ElectroStatic Discharge
HBM	Human Body Model

14. Revision History

Table 10. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS4321 Rev. 1.0	Jan 21, 2026	Product datasheet		